



Effect of mold pressure on compaction and ion conductivity of all-solid-state batteries revealed by the discrete element method

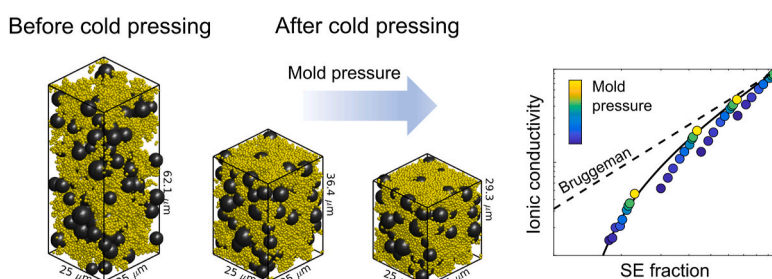
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HIGHLIGHTS

- We simulated a battery electrode using the discrete element method.
- In the cold press, we simulated aggregate destruction and particle consolidation.
- The highest stresses build up in the less ductile active material particles.
- Aggregates from ball milling cause a significant ionic percolation threshold.

GRAPHICAL ABSTRACT



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ABSTRACT

All-solid-state batteries (ASSBs) have enabled the development of compact and safer batteries with solid and non-flammable electrolytes. However, their practical applications are limited due to the difficulty associated with solid electrolytes (SEs) in forming a connected network for ionic transport and sufficient contact area on the active material (AM). Lithium-phosphate sulfides (LPS) are promising materials for SEs in ASSBs because they can plastically deform and form a close contact with AM. Herein, we develop and numerically implement a discrete element model for cold pressing that considers plastic deformation. Particle compaction proceeds in three steps, i.e., breakage of aggregates, particle rearrangement, and particle consolidation by plastic deformation. In addition, localized stresses easily deform the SE particles, thereby resulting in a force concentration between the AM particles. A higher mold pressure increases the relative density and contact area between the AM and SE. The relative density and ionic conductivity results agree well with the experimental results. We demonstrate a new correlation for ionic conductivity based on percolation theory from our observation of poor connectivity between SE aggregates. We believe that this research will help us understand the interactions during fabrication that can guide the development of future ASSBs.

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1. Introduction

1.1. ASSB advantages and challenges

Recent developments in Li-ion batteries (LIBs) have enabled electric vehicles to become viable alternatives to conventional vehicles driven by combustion engines. However, the energy density, which is still comparatively low, needs to be increased to meet consumer demand. The organic electrolyte is replaced by an inorganic solid electrolyte (SE) in all-solid-state batteries (ASSBs). SEs have many advantages: they are nonvolatile, non-flammable, leak-proof; have a wider voltage window; and can significantly improve battery energy density [1]. Therefore, they can help in the development of safer and more stable batteries.

Recently, Samsung developed a prototype pouch cell with a capacity of 800 km and a duration of over 1000 charges [2]. Toyota recently announced that their new battery will be released in 2021, which has a capacity of 500 km and allows for 10 min of fast charging. Toyota also expects to be the first company to release an ASSB-powered car in the 2020s [3].

Despite these announcements, there are several hurdles that must be crossed before ASSBs are widely available. SEs may have some disadvantages, such as low Coulombic efficiency (CE) and unstable cycling stability due to resistance at various solid–solid interfaces [4]. Because the electrolyte is in solid particle form, it is difficult to completely cover the electrode. In addition, significant SE–SE resistance may emerge owing to the grain boundary resistance caused by the misalignment of the crystalline structure between the grains [5].

1.2. Role of sulfide electrolytes on electrode compaction

There are several promising materials for SE with inherent electrical conductivities approaching those of liquid electrolytes. Numerous studies have attempted to develop materials with suitable chemical and electrochemical properties. Recently, the mechanical properties of SEs have gained more attention as they affect the growth of lithium dendrites, crack propagation, and interface resistance [6]. To reduce the tortuosity factor and constriction of the diffusive paths due to voids and cracks, the solid electrolyte must be densely packed. Dense SE layers have also been found to greatly suppress dendrite growth, especially for Li anodes [7].

Sulfides are mechanically compliant materials for ASSBs owing to their low hardness and low Young's modulus of elasticity [8], making them easy to deform and form a tight and connected distribution with a high contact area with the AM [9]. In this way, long-lasting and high-capacity ASSBs can be produced, such as Samsung's recent ASSB with an argyrodite-type sulfide electrolyte [2].

Amorphous lithium-phosphate sulfides (LPS or $\text{Li}_2\text{S-P}_2\text{S}_5$) are ductile groups of sulfides (Pugh's ratio ~ 2.4 [10]) that can be used to fabricate dense electrodes [11,12]. Ball milling and amorphization can be used to produce fine particle powders of LPS [13]. Further mixing and annealing can result in crystalized glass phases [14]. From these particles, a compact electrode can be produced by cold pressing at room temperature. Because of plastic deformation, the grain boundaries between SE particles are eliminated, and the transport properties are similar to those of hot-pressed samples [11,12]. However, wet chemical methods of LPS, which are to be used instead of ball-milling, have recently gained attention because of their scalability potential and the ability to obtain small evenly distributed SE particles, which may enhance ionic conduction [15,16].

1.3. Effect of electrode structure on the ionic conduction in ASSBs

The ionic transfer performance is influenced by the local electrode structure, from which the volume fraction of SE, φ_{SE} , and tortuosity factor of SE, τ_{SE} , can be derived. The tortuosity factor is a measure of the increment in the conduction time due to the curvatures in the con-

ducting path. The effective ionic conductivity of SE, $\sigma_{\text{SE}}^{\text{eff}}$, through the porous electrode is related to the relative conductivity, $\sigma_{\text{SE}}^{\text{rel}}$, and bulk conductivity $\sigma_{\text{SE}}^{\text{bulk}}$, as follows:

$$\sigma_{\text{SE}}^{\text{eff}} = \sigma_{\text{SE}}^{\text{rel}} \sigma_{\text{SE}}^{\text{bulk}} = \frac{\varphi_{\text{SE}} \sigma_{\text{SE}}^{\text{bulk}}}{\tau_{\text{SE}}}, \quad (1)$$

where φ_{SE} is related with AM fraction, f_{AM} , as follows:

$$\varphi_{\text{SE}} = (1 - \varepsilon)(1 - f_{\text{AM}}), \quad (2)$$

where ε is porosity. From the experiments on porous electrodes, it is common to fit the relation between the relative conductivity as a power relationship with the volume fraction of the conducting medium as follows:

$$\sigma_{\text{SE}}^{\text{rel}} = \frac{\varphi_{\text{SE}}^\alpha}{\gamma}, \quad (3)$$

where γ is the reduction factor and α is the exponent. In porous media applications, the Bruggeman relation is often applied, where $\gamma = 1$ and $\alpha = 1.5$. In the case of battery electrodes, the Bruggeman relation does not often provide a satisfactory fit, especially when the φ_{SE} is low. Therefore, a higher value of α is used to fit the data, usually within the range of $2.2 < \alpha < 3.6$ [17,18]; this is often referred to as the modified Bruggeman relation. However, even with this modification, the fit is sometimes poor at low volume fractions, and a composite function is necessary to fit both the high and small values of φ_{SE} [17].

Several studies have been conducted on the electrical conductivity of porous electrodes using the percolation theory [18,19]. When the particle volume fraction is low, the conductivity may drop to zero if the number of neighboring particles is insufficient to form a connecting chain. These studies suggest that there is a value of φ_{SE} above zero where the conductivity becomes zero, which may explain the difficulty associated with fitting the relative ionic conductivity with a modified Bruggeman relation.

1.4. DEM simulation and its significance for ASSBs

As discussed above, the mechanical properties of SE are important for densification [11,12] and the operating performance of ASSBs [6]. When compared to experiments alone, electrode reconstruction modeling allows for a more in-depth understanding of the effects of mechanical properties. The microscale structure of the battery electrode can be reconstructed using several top-down approaches: focused ion beam-scanning electron microscopy (FIB-SEM) analysis [20], X-ray computed tomography [21], time-lapse X-ray imaging [22] and operando visualization [23]. In particular, the latter techniques provide an accurate representation of the electrode morphology in both space and time. However, to better understand the effects of the mechanical properties on the morphology of the electrode, bottom-up models are preferred, such as molecular dynamics (MD) [5], coarse-grained molecular dynamics (CGMD) [24] and discrete element method (DEM) [24–27]. The discrete element method enables the simulation of individual particles and their interactions and can solve stress localization by particle misalignment directly. This is ideal for ASSBs because they are built on individual compressed grains.

The most fundamental DEM models have been applied to rock dynamics, fluidized beds and particle reactors [28]. However, by including hysteresis effects, plastic deformations in powder compaction can also be simulated [29]. Several studies have used the DEM for LIBs. They tend to focus on compaction during the calendaring process [24,30,31]. In previous studies, AM particles were simulated, and binder interaction was modeled implicitly as an interaction force [30,31]. In addition, LIB fabrication has been simulated in three steps with an explicit description of the AM and carbon-binder domains [24], wherein the first two steps involve the CGMD of the slurry and the drying steps, whereas in the final

step, calendaring of the electrode was incorporated into a DEM simulator.

1.5. Scope of this study

In this study, we investigated the effect of the cold-pressing pressure and fraction of AM on the mass transfer performance. We performed three types of simulations in the following order: (i) annealing, (ii) aggregate settling, and (iii) cold pressing. The latter is the core of this study, and we developed a novel 3D DEM model in MATLAB that includes particle consolidation in a mold in room temperature. The plastic deformation model was based on a previous two-dimensional model [27]. In this study, we extended it into three dimensions, which we concluded was necessary to increase the accuracy of the porosity and mass transfer calculation. First, before the actual cold-pressing simulation, we conducted an annealing simulation that mimics the ball-milling, and annealing processes to produce aggregates. Second, we conducted a simulation in which gravitationally packs the aggregates in the cell before cold pressing.

Finally, after cold pressing, the tortuosity factor was calculated from which the relative ionic conduction coefficient was estimated. If the grain boundaries are significant, this method overestimates mass transfer. However, if Sakuda's hypothesis is correct, most grain boundaries have vanished due to plastic deformation at the particle contact points [11,12], and the above method will be accurate because the relative ionic conductivity will only depend on porosity and the tortuosity factor. The DEM methods have unique features that can elucidate (i) the dominant interaction forces, (ii) the region of stress localization, and (iii) the effect of structures on mass transfer performance, all of which are important for developing future ASSBs.

2. Methods

2.1. Simulation strategy

We reconstructed various SE layers and electrodes with different mold pressures. We carried out simulations with different fractions of AM, f_{AM} , of which the SE layer correspond to the simulation without AM, $f_{AM} = 0$. The simulation was conducted in three steps, i.e., aggregation, settling, and cold pressing. The cold pressing in turn included a pressing and release time period.

2.2. Simulation of annealing-induced aggregation

An aggregate construction model of SE particles was developed to mimic the interactions occurring during the annealing step after the ball milling process [14]. This model was initially developed for the annealing of carbon black in fuel cells in our group, but the principal annealing process was similar [32,33]. Aggregates were created from an initial fixed particle and a free particle, where the latter performs a random walk until it attaches to the surface of the fixed particle. The random walk represents mixing and diffusion, whereas attachment represents the formation of fusion bonding between particles. An aggregate is eventually formed by the subsequent addition of random walkers. Because of the higher likelihood of particles attaching to the convex parts of the aggregates, the morphological structure of the aggregates becomes branch-like. The simulation proceeds until a desired aggregate size of 20 μm is reached. Overlaps between the particles were controlled by a shoving algorithm [34] with a maximum overlap distance of 0.4 R_{eff} .

2.3. DEM simulation of aggregate settling

This part of the simulation represents aggregate packing in the cell after the ball milling and annealing processes. Initially, the aggregates were distributed far apart to avoid overlap, but gradually, they settled to

the bottom. In this part of the simulation, the particles within each aggregate are assumed to be in force balance and only exert forces with particles of other aggregates. The aggregates were assumed to be rigid and only allowed to move by translational and rotational movements by Newton's laws of motion. This part of the simulation was previously developed for the settling of aggregates/agglomerates in fuel cell catalyst ink [32]. The compressive forces were low; therefore, plastic deformation was neglected in this part of the simulation. The contact force interactions are explained in detail in section 2.5.

2.4. DEM simulation of cold pressing

The DEM cold-pressing simulation (room temperature) starts off where the settling simulation ends, but with a moving wall added to the top. The top wall moves by controlling the contact pressure with the nearby particles. The pressure of the wall gradually increases until it reaches the desired mold pressure, which is controlled by a numerical proportional-integral (PI) controller. The bottom wall was fixed and periodic boundary conditions were applied in the x and y directions. First, any particle existing any side along x or y will reappear on the opposite side. Additionally, ghost particles were added outside the periodic boundaries by copying particles close to the opposite side. In this part of the simulation, the compressive stresses are sufficiently large to cause plastic deformation. The aggregates were allowed to deform if the stresses between the particles exceeded the hardness. In addition, new fusion bonding between particles will develop if the compressive stresses are sufficiently large. This model was developed in a previous study in two dimensions [27] and has been extended to three dimensions in this study. A scheme of our interaction model between two particles, i and j , is shown in Fig. 1. Cundall and Strack [35] developed the initial soft-sphere model, which is represented by the sketch in (a), excluding the blue dashpot. It includes a spring and a dashpot in both the normal and tangential directions and an additional slider in the tangential direction. The contact force is calculated based on the overlap distance of two contacting spheres h_{ov} , as shown in (b). Our previously developed model [27] is based on a nonlinear Hertzian spring [36] for elastic interaction and a hysteresis model for plastic deformation. The plastic deformation model is illustrated using a blue dashpot in (a). In addition, because of the deformation, the distance for which the particles are in contact is reduced. We denote this change by the equilibrium overlap h_{eq} , which is the overlap of the representing spheres for which the contact forces are zero, as illustrated in (c). The parameters used in the simulations are listed in Table 1. Details of the particle contact interactions are explained in sections 2.6-2.7.

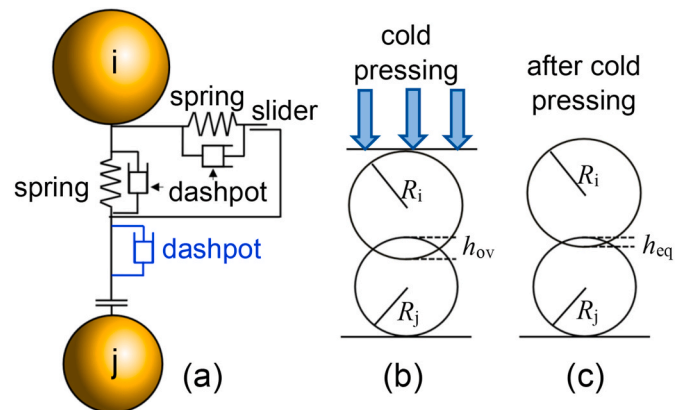


Fig. 1. Schematics of our DEM model. (a) Particle interactions with an added dashpot (blue) for plastic deformation. Relation between (b) the overlap distance h_{ov} during cold pressing and (c) the equilibrium overlap distance h_{eq} after cold pressing. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 1
Parameters in the simulation.

Symbol	Value	Units	Description	Reference
E	0.5	–	Restitution coefficient	This study
E_{SE}	24	GPa	SE Young modulus (LPS)	Sakuda et al. [12]
E_{AM}	70	GPa	AM Young modulus (Si)	Sethuraman [37]
μ	0.5	–	Friction coefficient	This study
ν	0.3	–	Poisson ratio	This study
σ_H^{AM}	10.6	GPa	Hardness of AM (Si)	de Vasconcelos et al. [38]
σ_H^{SE}	1.9	GPa	Hardness of SE (LPS)	McGrogan et al. [8]

2.5. Analysis and estimation of mass transfer properties

After the DEM simulation, the tortuosity factor, relative diffusion coefficient, and contact area were estimated. First, the volume fractions were mapped onto a 3D grid. This mapping was performed by first finding the grid points at a distance within a particle radius to its center. However, because of plastic deformation, this would have underestimated the volume fraction. To obtain the correct volume fraction, the threshold distance had to be recalculated and iterated until the correct volume fraction within a 0.1% relative tolerance was obtained. The SE tortuosity factor and relative diffusion coefficient were estimated by an external MATLAB application *TauFactor* developed by Cooper et al. [39]. After correspondence with the authors, a pre-release version 2, which includes periodic boundary conditions, was provided and used in our calculations.

2.6. Governing equations for the DEM simulations

In the DEM simulations, we solved for the Newtons law of motion, where particles accelerate with the response to applied forces. The contact force in the normal direction is given by

$$F_n = (-F_{\text{spring}} + \eta(u_i - u_j) \cdot n)n, \quad (4)$$

where F_{spring} is the spring force. In the aggregate settling simulation, where plastic deformation was neglected, was given by

$$F_{\text{spring}} = k_n h_{ov}. \quad (5)$$

Here, k_n [N m^{-1}] is the normal spring coefficient and h_{ov} is the overlap distance indicated in (b). Our model is based on a nonlinear Hertzian spring [36] for which the spring stiffness is a function of h_{ov} is given by

$$k_n = 4/3 \cdot E_{\text{eff}} h_{ov}^{1/2} R_{\text{eff}}^{1/2}, \quad (6)$$

where R_{eff} [m] is the effective radius based on neighboring particle radii R_1 and R_2 [m]

$$R_{\text{eff}}^{-1} = R_1^{-1} + R_2^{-1}. \quad (7)$$

This relation can also be used for particle-wall interaction by setting the radius of the wall to infinity. E_{eff} [Pa] is the effective elastic modulus given by

$$E_{\text{eff}}^{-1} = (1 - \nu_i^2)E_i^{-1} + (1 - \nu_j^2)E_j^{-1}. \quad (8)$$

Here, E_i , E_j [Pa] are the Young moduli and ν_i and ν_j [–] are the Poisson ratios. The damping coefficient η [35] is a function of the restitution coefficient e (Table 1) and is given by

$$\eta = -2\ln(e) \sqrt{\frac{m_{\text{eff}} k_n}{\ln^2(e) + \pi}}, \quad (9)$$

where m_{eff}^{-1} is the effective mass

$$m_{\text{eff}}^{-1} = m_i^{-1} + m_j^{-1} \quad (10)$$

The model also includes tangential friction F_t , which has a limit before slippage according to the Coulomb friction model:

$$|F_t| \leq \mu |F_n| \quad (11)$$

where the right-hand side is the kinetic friction and μ is the friction coefficient.

2.7. Adaptation of DEM simulation with plastic deformation

In our new DEM simulation of cold-pressing, we recently started to develop [27], included plastic deformation, as indicated by the blue dashpot in Fig. 1 (b). The normal force is expressed with an equilibrium overlap h_{eq} , which appears as a result of the new position of the plastically deformed non-spherical surface at the contact point between particle pairs. The spring force yields the following:

$$F_{\text{spring}} = k_n (h_{ov} - h_{eq}) \quad (12)$$

and the relation of k_h with h_{ov} is maintained as in equation (3) for intermediate deformations. The rate of change of the overlap is modeled by the following relation:

$$\frac{dh_{eq}}{dt} = \frac{F_{\text{spring}} - F_{th}}{t_{rel} k_n} \quad (13)$$

where t_{rel} is the relaxation time and F_{th} is the threshold force.

$$F_{th} = 2/3 \cdot \min(H_i, H_j) A_{con} \quad (14)$$

where the factor 2/3 is the ratio of the average to maximum stress with the assumption of a parabolic stress profile. Here, H_i , H_j are the hardness of the particle materials and A_{con} is the contact area:

$$A_{con} = \pi h_{ov} R_{eff} \quad (15)$$

2.8. DEM implementation in MATLAB

All simulations were conducted using the MATLAB® code developed in our laboratory. The equations described in previous section was implemented with second order accurate Verlet scheme for the time integration [40]. Because of the large number of particles (45,000), the brute-force method of searching for all particle contacts with all other particles would lead to unacceptably long simulation times. Instead, we used a kd-tree search algorithm [41] to calculate the nearest neighbors on a regular basis. It is a type of space partitioning tree for organizing points in boxes to obtain the closest neighbors efficiently. However, because the algorithm checks for the Euclidian (center-center) distance, it may cause problems when the sizes of AM and SE are significantly different. Therefore, it was better to group the particles according to their type before the search. The neighbor search was then processed by checking for the AM-AM, AM-SE, and SE-SE group combinations separately. The number of searched neighbors was 20, 150, and 20. AM-SE is roughly based on the number of SE particles that can surround a single AM particle. In MATLAB, the kd-tree algorithm was accessed through statistics and machine learning toolbox using the commands “knnsearch” and “createns”. We ran the simulations on a workstation with an Intel® Xeon® W-2145 CPU (3.70 GHz, 8 cores) and each simulation took approximately 25 min.

3. Results

3.1. Particle distribution and morphology

The spatial distribution of the particles obtained from the simulation of the SE layer is shown in Fig. 2, (a) before and (b) after aggregate settling. The particle distribution of the SE layer (c, d) during and (e, f) after cold pressing can be observed at 100 MPa and 400 MPa. Note that

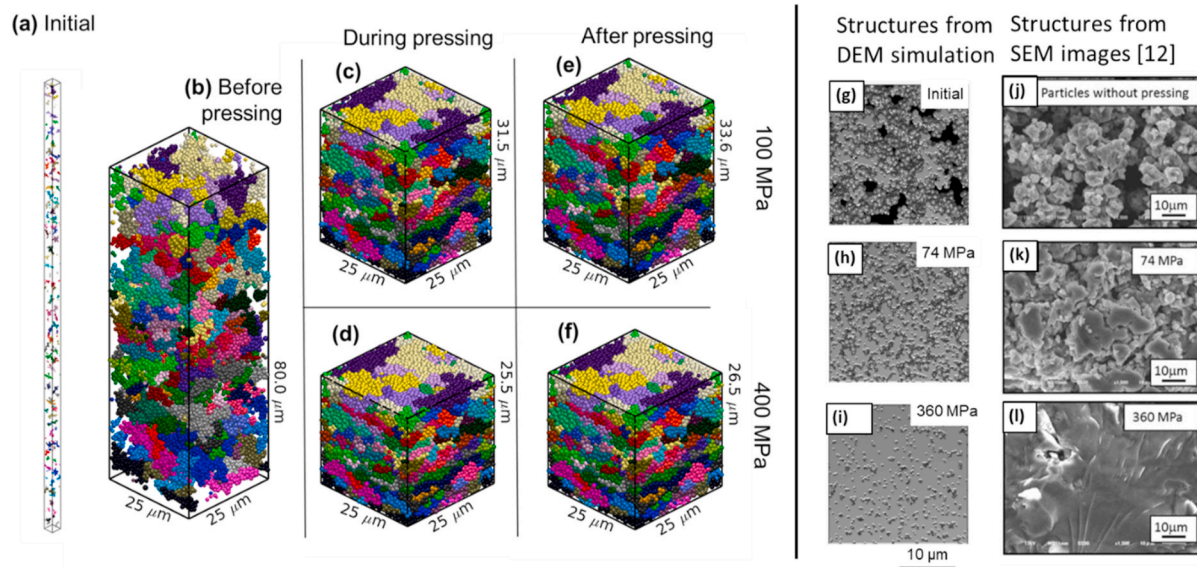


Fig. 2. Spatial particle distribution of particle structures in SE layers initial, before, during and after cold-pressing. The SEM images (j–l) are reproduced from Sakuda et al. [12], with open access permission by Springer Nature.

the height of the electrode expands slightly when the pressing stops. Comparisons of the structures between DEM (g–i) and the scanning electron microscopy SEM images of Sakuda et al. [12] can be observed (g, j) before and after cold-pressing at (h, k) 74 MPa and (i, l) 360 MPa.

The particle positions, including AM particles, are shown in Fig. 3 with a 50% solid volume percentage of AM. For the SE layer, the anode layer becomes more compressed with higher pressure, and the electrode slightly expands upon release.

3.2. Effect of mold pressure and composition on compaction

The structural properties of the SE and anode layers are shown in

Fig. 4. The time series of the electrode height is shown in (a). The higher the applied pressure, the more compressed the final SE layer thickness. The relative density of the final SE layer is plotted with the mold pressure in (b), and the simulations are compared with the experimental results of Sakuda et al. [12]. The effect of the AM fraction in the electrode on the relative density is shown in (c). The higher the AM fraction, the lower the final density of the electrode. The SE coverage of AM is plotted in (d). The higher the AM fraction, the lower is the SE coverage on the available AM surface.

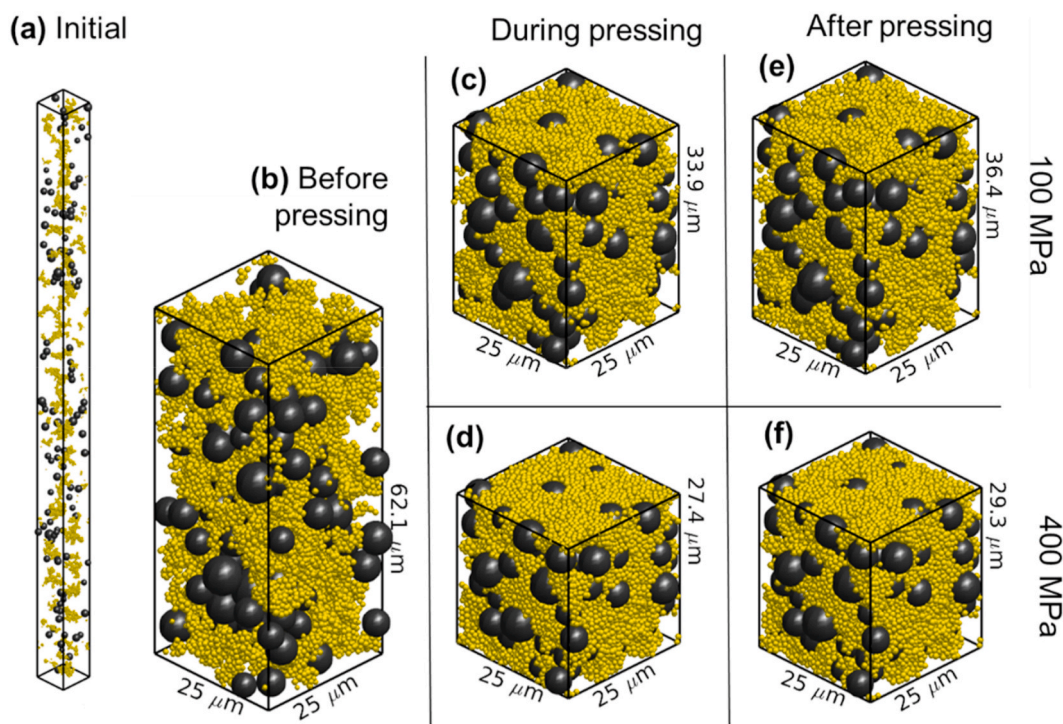


Fig. 3. 3D particle distribution plot of an anode electrode including AM particles ($f_{AM} = 0.5$).

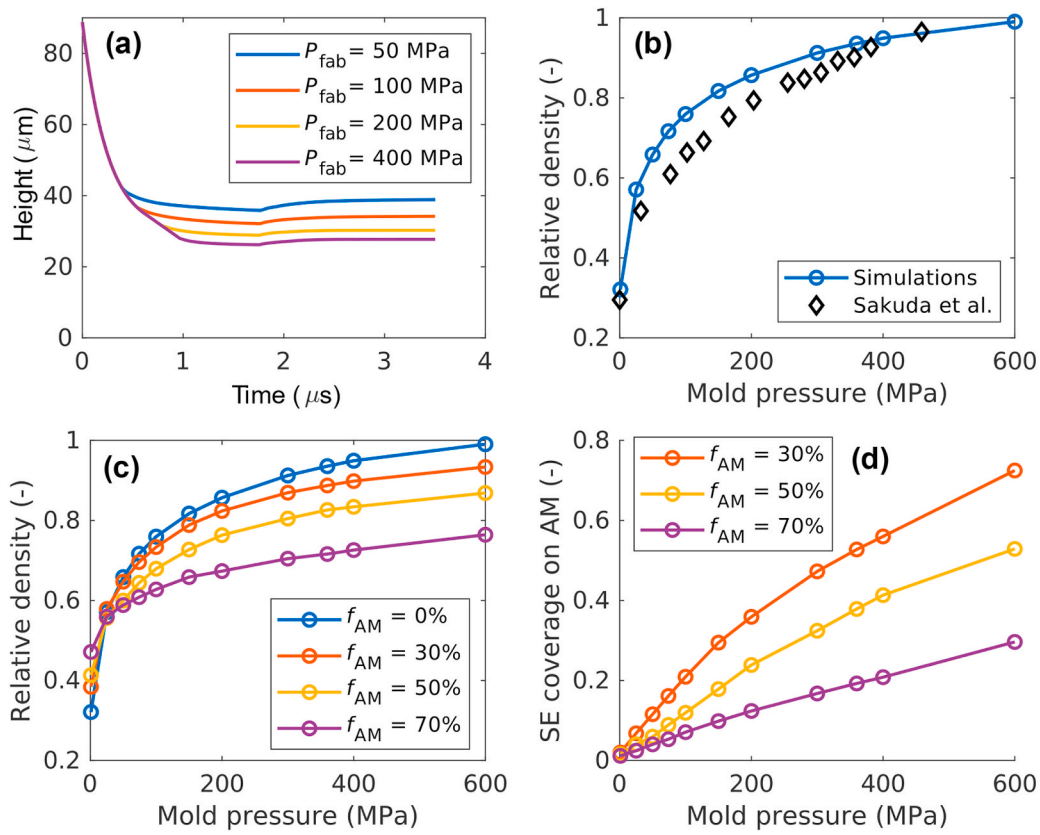


Fig. 4. Structural properties of the SE-layer and the anode electrode.

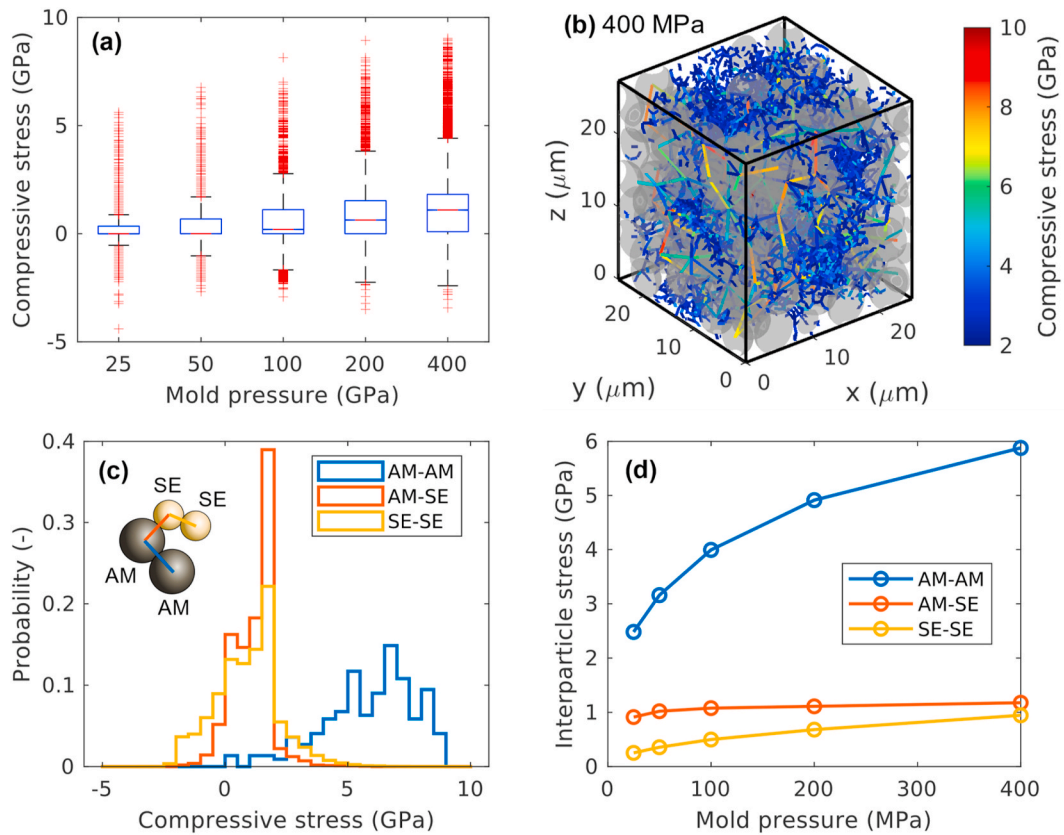


Fig. 5. Compressive stress distribution of the anode layer illustrated by a box plot for different mold pressures (a), 3D plot at 400 MPa (b), histograms of different types of particle pairs (c), and plots with mold pressure (d).

3.3. Effect of mold pressure and composition on internal stresses

Various plots of compressive stress distributions are shown in Fig. 5. A boxplot of the compressive stresses at different mold pressures is presented in (a). The boxes contain 50% of the data with boundaries in the first and third quartiles of the distribution. The whiskers extend to a distance of 1.5 times the height of the box, and the red crosses are the outliers. The median value of the compressive stress, indicated by the red lines inside the boxes, increased significantly with mold pressure. However, there is a wide distribution of stress magnitudes that needs to be investigated in detail. The stress network between the particles at a mold pressure of 400 MPa is shown in (b). For visibility, only the AM particles and the largest stresses between the particles are shown. In addition, a network of high stresses is visible between the AM particles, which are indicated by the longer red and yellow lines. There is also a separate network of low stresses between the SE particles, which is denoted by the shorter blue lines. To confirm this observation, three histograms are plotted in (c), where each histogram resembles the distribution for each particle-pair type, i.e., AM-AM, AM-SE, and SE-SE (see illustration). The stress distribution at the AM-AM contacts is far greater than that at both the AM-SE and SE-SE. In addition, the average stresses for different particle-pair types are plotted with the mold pressure in (d), and the stress concentration at the AM-AM contacts is consistently dominant at all mold pressures.

Fig. 6 shows the degrees of compression and plastic deformation. We defined the equilibrium overlap ratio as $h_{ov}/4R_{eff}$. For particles of similar size, it is defined as the relative overlap divided by the sum of the diameters. This is a measurement of fusion bonding on the aggregates and cold-pressed particle structures. At the start, there are preexisting cohesive contacts between particles in the aggregates developed during annealing in the ball-milling process. Therefore, the equilibrium overlap ratio is not initially zero. A box plot of the equilibrium overlap ratio is shown in Fig. 6 (a). The median value increases with mold pressure; however, there is a wide distribution of values. In Fig. 6 (b), the values have been averaged over different types of particle pairs. For small mold pressures, the largest values can be found for SE-SE contacts, which indicate fusion bonds within aggregates. When the mold pressure increases, a significant increase in the values for AM-SE contacts can be observed, which indicates the lamination of SE particles on the surface on AM.

3.4. Effect of compaction on ion-conductivity

The ionic transfer properties of the fabricated electrodes are shown in Fig. 7 and compared with the experimental results of Sakuda et al. [12] ($f_{AM} = 0$) and Nakamura et al. [42] ($f_{AM} = 0.742$). In (a), the relative ionic conduction coefficient of the SE layer is plotted with the

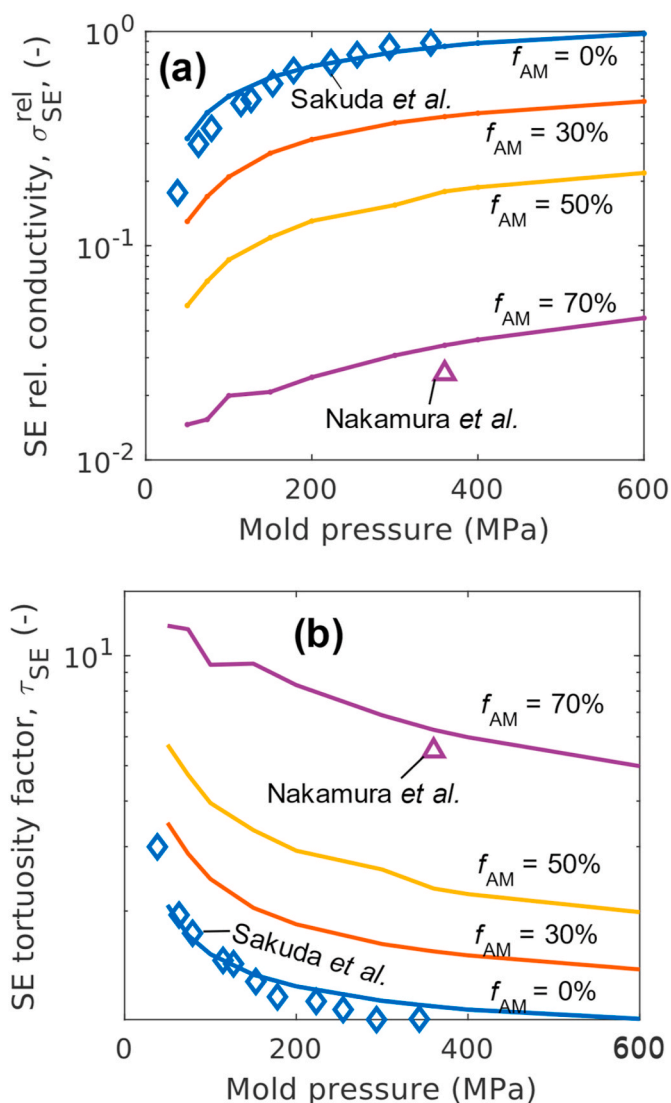


Fig. 7. Ionic transfer properties in SE plotted with mold pressure and comparison with experiment of Sakuda et al. [12] and Nakamura et al. [42]. (a) Relative conduction coefficient and (b) tortuosity factors are plotted with different AM fractions.

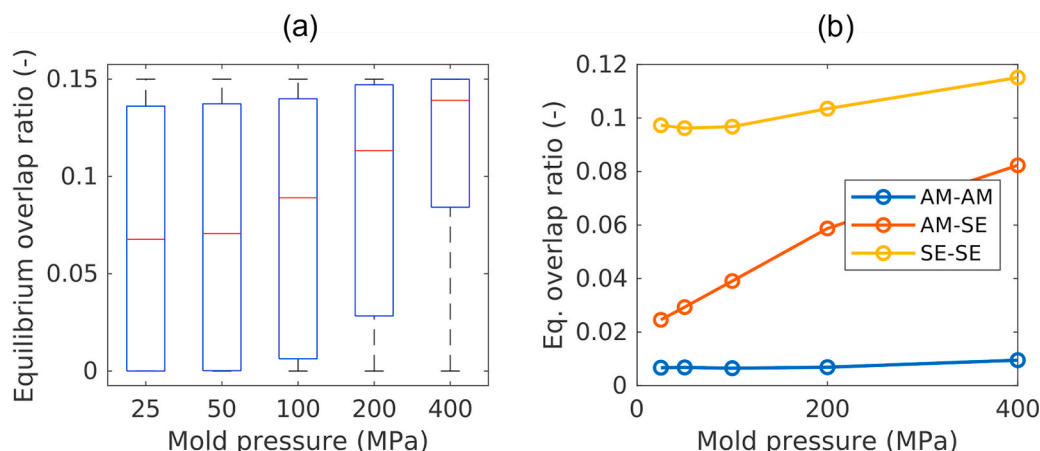


Fig. 6. Effect of mold pressure on the equilibrium overlap ratio, presented in a box plot (a) and average values for different particle pairs (b).

mold pressure. The ionic conductivity increases with mold pressure and reduces with f_{AM} . Reversed trends are shown for the tortuosity factor in (b) which decrease with mold pressure and increase with f_{AM} . The results of these ionic transfer properties agree well with the experimental results. Although the study of Nakamura et al. focused on SE coated AM, the data is taken from their reference experiment on conventional random mixture that closer resemble our DEM simulations.

3.5. Effect of compaction on ion-conductivity

The relationship between the SE volume fraction and the relative ionic conduction coefficient, as obtained through our simulations, is shown in Fig. 8. In Fig. 8 (a), the points from our simulations are mapped and colored by the mold pressure. The Bruggeman correlation is indicated by a dashed line, which shows a reasonable agreement at a high value of ϕ_{SE} for higher fabrication pressures; however, the relation deviates further away at lower values of ϕ_{SE} . Furthermore, the points at high fabrication pressure do not follow a straight line in the log-log plot, which means that it is impossible to obtain a good fit with a modified Bruggeman relation with an increased exponent; the points tend to dip rapidly below a certain value of ϕ_{SE} . This drop can be explained by the percolation theory [19]. Based on this consideration, we developed a new relationship for ionic conduction in SE as follows:

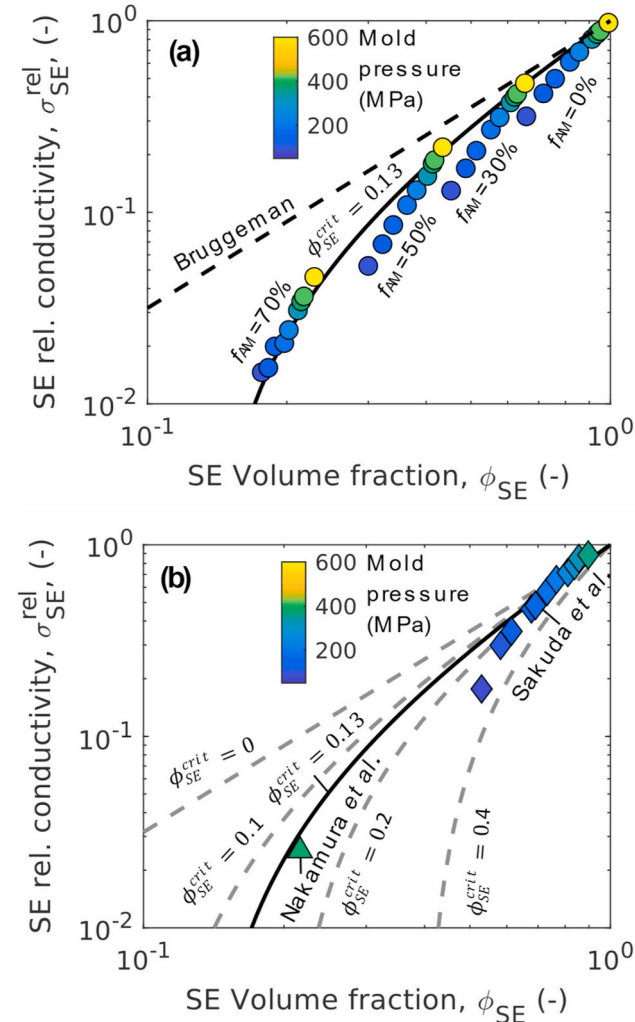


Fig. 8. Relative conduction coefficient and the dependence of the SE volume fraction. (a) Comparison between our simulations and our new correlation. (b) Comparison between our new correlation with experiments of Sakuda et al. [12] and Nakamura et al. [42].

$$\sigma_{SE}^{rel} = \left(\frac{\phi_{SE} - \phi_{SE}^{crit}}{1 - \phi_{SE}^{crit}} \right)^{\frac{1}{2}}, \quad (16)$$

where ϕ_{SE}^{crit} is the critical value of ϕ_{SE} below which the ionic conductivity becomes zero. The black line corresponds to a value of $\phi_{SE}^{crit} = 0.13$, and this value was fitted in the range of common mold pressures (200–400 MPa) during fabrication, corresponding to the green colored circles in Fig. 8 (a). With this empirical formula, it is possible to obtain a good fit with the experimental data without the use of a composite function. In addition, when ϕ_{SE}^{crit} is set to zero, the correlation becomes the Bruggeman relation. In Fig. 8 (b), previously reported experimental data are added [12,42], and the effect of varying ϕ_{SE}^{crit} can be seen.

4. Discussion

4.1. Novelty and importance of this study

In this study, we conducted three simulations: (i) annealing, (ii) aggregate settling, and (iii) cold pressing. In comparison to earlier studies of ASSB using DEM, this study includes the plastic deformation of particles during pressing and release. This novel implementation in the DEM simulation of ASSBs makes stress predictions in cold pressing more realistic. Between SE particles, where plastic deformation occurs, stresses are reduced, whereas between AM particles, which experience minor plastic deformation, stresses are magnified. The simulations can predict the packing of the cell and perform the subsequent mass transfer prediction. Therefore, this study can be a valuable tool to aid in the design and fabrication of future ASSBs.

4.2. Summary of the results

The particle physics in the cold-pressing stage was found to be a complex interaction of aggregate breakage in the initial stage of the simulation, followed by particle rearrangement and particle consolidation by plastic deformation.

Because of the plastic deformation of the SE, the stresses between the SE particles were dampened as cold pressing proceeded. The same were not observed for AM particles because of their higher hardness. The AM particles are mostly deformed elastically without plastic deformation, which causes stress to build up between the AM-AM particles. This observation should be considered to avoid the destruction of AM particles during fabrication with cold pressing.

Our simulation results of cold pressing using annealed aggregated SE particles agree well with those of Sakuda et al. [12] both in terms of relative density and relative ionic conduction coefficient. The relative conduction coefficient could be estimated based solely on the geometry (porosity and tortuosity factor) without including the effects of the SE-SE grain boundary resistance.

The relationship between SE fraction was compared with the tortuosity factor and relative conduction coefficient. The results of relative conduction revealed that the Bruggeman correlation significantly overestimated the relative conduction coefficient. This can be explained by the aggregation of the SE particles, which increase the tortuosity factor in the location between aggregates.

4.3. Shortcomings and future research

There are various ways that this model can be used in future studies, for example: (i) the effect of charge/discharge on durability and performance, (ii) development of a viscoelasticity model of Li to simulate the growth of dendrites, and the (iii) implementation of coupling with MD or CGMD. The same model can be used for simulating charge/discharge, but with different boundary conditions. The particle expands during charging, and the wall pressure is maintained at a stack pressure,

which may increase with the intercalation expansion. By slightly modifying the model structure, the model can be used to simulate viscoelastic materials, such as Li metal. This can be a valuable tool to simulate ASSBs with Li anodes and to simulate the growth of dendrites, which can also deform with applied stress. Similar to the previous research on LIBs [24], DEM can be coupled with MD to upscale molecular interactions, and speed-up can be achieved by particle-swarm optimization methods [43].

In our simulations, AM was modeled as spherical particles. However, it may be necessary to simulate non-spherical particles in a future model. Moreover, the current model layout with single AM particles cannot simulate intraparticle cracking within AM particles. In a future model, we plan to simulate AM particles as clusters or aggregates of smaller particles. Thus, the non-spherical shape of AM and intraparticle cracking of AM can be resolved. Furthermore, it would be possible to find an optimum fabrication pressure that would result in a sufficient electrode density, but not high enough to cause significant AM cracking.

5. Conclusions

From various particle simulations, this list summarizes the advances in knowledge that emerged from this study:

- The compaction of a solid-state battery electrode is a complex interaction of aggregate destruction, particle rearrangement, and particle consolidation.
- Our DEM simulation using annealed aggregates of SE particles produced results that were in close agreement with previous experimental results of the relative density. The relative ionic conduction coefficient obtained from the tortuosity factor calculations also agreed well with the experiments without adding grain boundary resistances between the grains.
- Owing to plastic deformation, the stress is reduced at the SE–SE and AM–SE particle contacts. However, because AM material is harder and does not deform plastically to the same extent, substantial buildup of stresses occurs at AM–AM contacts.
- Owing to the aggregation of the SE particles after the ball-milling process, isolated regions in SE developed to a greater degree than those developed in random free particles. Therefore, the Bruggeman correlation does not accurately predict the tortuosity factor and the relative conductivity of the SE phase. A new correlation for SE conduction in ASSBs considers the lack of connectivity attributed to the aggregated ball-milled SE particles, and it includes a critical value below which the ionic conductivity is zero.
- The DEM simulation model developed was demonstrated to be effective to predict the ionic performance and is demonstrated to be a valuable tool to aid in the design and fabrication of future ASSBs.

CRedit authors contribution statement

Magnus So: Conceptualization, Writing, Methodology, Software, Formal analysis. **Gen Inoue:** Conceptualization, Supervision, Funding acquisition. **Ryusei Hirate:** Investigation. **Keita Nunoshita:** Investigation. **Shota Ishikawa:** Investigation. **Yoshifumi Tsuge:** Project administration, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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